

QUEST: From Data to Decisions

Module 1 — Foundations: Critical Data Literacy, Explainable AI, and Data
Storytelling with R

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July 10, 2026

ICOTS-12 Pre-Conference Workshop · Brisbane, Australia

One Case Study, One Journey

Over the next few hours, we are going to walk through **one complete case study** together.

Using a real educational dataset on **student Mathematics and Portuguese performance**, we will investigate one fundamental question:

Why do some students succeed while others struggle?

One Dataset. One Question. One Complete Analytical Journey.

Over the next few hours, we will work together on a single educational case study.

Instead of treating statistics, machine learning, and artificial intelligence as separate topics, we will integrate them into one coherent workflow that moves from **questions** to **evidence** and ultimately to **better educational decisions**.

By the end of this workshop, you will practice:

**Critical Data Literacy → Exploratory Data Analysis → Machine Learning →
Explainable AI → Data Storytelling**

Rather than learning isolated statistical techniques, we will follow the **QUEST Framework** to:

1. **Q – Question** the educational problem.
2. **U – Understand** the data through critical exploration.
3. **E – Explain** predictive models using Explainable AI.
4. **S – Story** the evidence using effective data storytelling.
5. **T – Transform** statistical evidence into responsible educational decisions.

Today's Schedule

Time	Session	QUEST
9:00 – 10:30	Session 1: Question & Understand	Q, U
10:30 – 11:00	<i>Morning Tea</i>	
11:00 – 12:30	Session 2: Explain	E
12:30 – 1:15	<i>Lunch</i>	
1:15 – 3:00	Session 3: Story & Transform	S, T

By the End of Today

- You will have questioned, explored, modelled, explained, and audited ONE real dataset, start to finish, as a single continuous investigation.
- You will know a simple framework, **QUEST**, that you can teach again, in your own classroom, with your own data.
- Every number you see today is real. Nothing is simulated for effect, and every claim can be re-checked from the code you'll leave with.

Data $\rightarrow$ Model $\rightarrow$ Accuracy

Collect data. Fit a model. Report how well it performed. Done.

What's Changed

- Machine learning models now make or inform real decisions about real people — who gets a loan, who gets flagged at school, who gets released on bail.
- Our students will use these tools, build them, or be affected by them, whether or not we teach them to.
- “The model was 85% accurate” is no longer a complete answer to anything that matters.

*Statistics education already teaches critical thinking about numbers.
That habit of mind now needs to extend to algorithms.*

This Is Not a New Subject

Everything in today's workshop builds on ideas you already teach:

Sampling and estimation	↔ train/test splitting
Confounding and bias	↔ proxy variables, fairness
Uncertainty	↔ calibration, confidence
Effect size	↔ feature importance

We are not replacing what you teach. We are extending it.

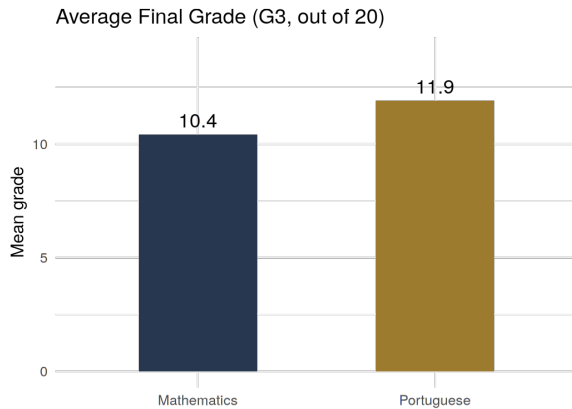


Figure 1: Math vs Portuguese Score

Why?

Take thirty seconds. What's your first explanation?

Several assumptions immediately come to mind:

Portuguese is easier than Maths.

Maths teachers grade harder.

Students who struggle in Maths give up sooner.

...

Notice: every one of those is a guess.

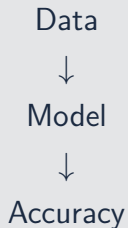
**We haven't looked
at the data yet!**

Data never speak for themselves.

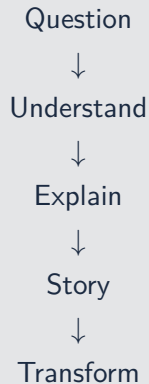
Every value in that data is real: 395 Mathematics students, 649 Portuguese students, same school district, same survey instrument. Everything else is a story someone tells about it.

A Different Workflow

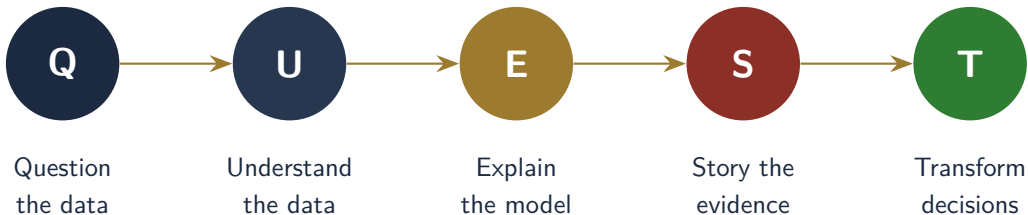
The old way



The QUEST way



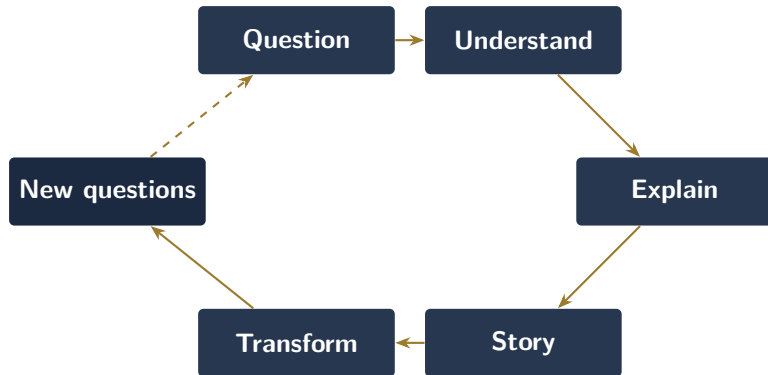
The QUEST Framework



What Each Letter Means

Q	Question the data	Critical data literacy — what does each column measure, and what does it stand in for?
U	Understand the data	Exploratory analysis — generate questions, not answers
E	Explain the model	Global and local explanation, from more than one method
S	Story the evidence	The same finding, told honestly, for different audiences
T	Transform decisions	A documented, defensible, evidence-based action

QUEST Is a Cycle, Not a Line



Every decision raises the next question. QUEST doesn't end — it loops.

Data Literacy vs. CRITICAL Data Literacy

Data literacy

Can you read a table, a chart, a summary statistic?

Critical data literacy

Can you question who collected it, why, and who's missing from it?

Today's dataset has 33 columns describing 395 real students. Before we model anything, we interrogate the columns themselves.

Every dataset has a history.

Before asking,

"What do the data tell us?"

we should first ask,

"Where did these data come from?"

Because critical data literacy begins before data analysis.

What's Missing in the Data?

- Household income
- Teacher quality
- Class size
- School funding
- Nutrition
- Student mental health
- Motivation
- Peer effects
- Home language
- Learning disabilities

A prediction can only be as good as its data.

Think Like a Critical Data Scientist

Before opening R, ask yourself... the 5Ws of data

1. Why were these data collected?

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1. Why were these data collected?
2. Who decided which variables to measure?
3. Which students are represented?
4. What might be missing?
5. Where? Can findings from only two schools represent an entire country?
6. Would these results generalise to
 - Brazil?
 - Nigeria?
 - South Africa?
 - Australia?

Key Message

Every dataset reflects the context, purpose and decisions made during its collection.

Four Kinds of Evidence

Tag	Category	Example
HF	Hard fact	age, school, failures
SR	Self-report	famrel, health, studytime
PX	Proxy	address, Medu, Fedu
SN	Sensitive	Dalc, Walc, romantic

A number is not a fact just because it's numeric.

When you look at the data, perform:

Exploratory data analysis: distributions, group comparisons, correlations etc.

**But EDA generates
more questions.**

Not answers.

A correlation between study time and grade is a lead, not a conclusion. This session (Session 1) spends real time here, before any model is fit.

Every plot tells a story.

*But does it tell the
whole story?*

A model can predict well and still teach you nothing about *why*.

**Accuracy is a
starting point,
not a destination.**

A Real Preview: The Accuracy Paradox

REAL RESULT

	Mathematics	Portuguese
Accuracy	63%	82%
Actual fails caught	24%	12%

Higher accuracy, **WORSE** at catching the students who actually fail. We'll see exactly why this afternoon.

It's:

Accurate for whom?
Wrong in which direction?
Trusted by whom, and why?

Can we trust the model used to predict student performance?

This is the whole point of this afternoon's session.

Global and Local Questions

Global explanation

What does this model generally pay attention to, across everyone?

Local explanation

Why did it make THIS prediction, for THIS one person?

A pattern that holds on average is not automatically true of any one individual.

REAL RESULT

One real student in today's data: 3 past failures- the strongest single warning sign in the whole dataset. The model was 87% confident this student would fail.

That student passed.

Three Independent Methods

This afternoon, we'll interrogate the SAME model three different ways:

`icarm` explanation built into the model-fitting call itself

`DALEX` a model-agnostic break-down, works on any model

`LIME` an entirely independent local-explanation method

Where independent methods agree, trust grows. Where they disagree, that's information too.

Same Results, Different Audiences

Imagine presenting the SAME finding to:

Teachers
Parents

School leaders
Students

Media
Policy makers

Same evidence. Different framings. Same obligation to honesty in every one.

Bad, Better, Best

Bad

Random Forest produced 68% accuracy.

Better

Students with repeated failures are at greatest risk of poor outcomes.

Best

Early academic support for students with previous failures may help more than focusing on attendance alone — failures outweighs every other variable, in every model family we tried.

*A number without a story
invites the listener
to invent one.*

This afternoon: the Four C's — a simple structure for making sure that story is honest.

Today's entire pipeline runs through one R package family:

`civic.icarm`



`icarm`

From Civic Education to General Practice

`civic.icarm`

Built first, for civic and political education, reproducible, fairness-aware analysis of survey and civic data.

`icarm`

Extends the same philosophy to ANY tabular data, health, finance, social science, and (today) education.

Same underlying commitment: interpretable, contextual, accountable, and responsible modelling — ICARM.

One Interface, Many Models

```
icarm_fit(pass ~ ., data, model = "cart")  
icarm_fit(pass ~ ., data, model = "random_forest")  
icarm_fit(pass ~ ., data, model = "svm")
```

Interpretable models and black-box models, through the exact same call.
Comparing them becomes a one-line decision, not a rewrite.

What icarm Gives You, Built In

- **Unified fitting** — interpretable and extended models, one interface
- **Explanation** — global importance and local, case-by-case reasoning
- **Fairness auditing** — group-level accuracy and error-rate comparison
- **Calibration** — are the predicted probabilities actually trustworthy?
- **Audit trails** — a timestamped, reproducible record of every decision

Not just another modeling package, it is a framework for responsible, reproducible data science.

What We've Covered

- Why statistics education needs to extend into explainable, responsible modelling
- One real result (Maths vs. Portuguese) that a story, not a number, actually explains
- The QUEST framework: Question, Understand, Explain, Story, Transform
- Why critical data literacy, EDA, explainability, and storytelling each matter on their own
- The tool we'll use all day: `icarm`, and its civic-education predecessor `civic.icarm`

**Now let's question
the data.**

Meet the Student Performance Dataset

Original Source

The Student Performance dataset was collected by

- **Paulo Cortez**
- **Alice Maria Gonçalves Silva**

at the **University of Minho, Portugal.**

The data were collected from

- two Portuguese secondary schools
 - Gabriel Pereira (GP)
 - Mousinho da Silveira (MS)
- school administrative records
- student questionnaires